

Тема

SQL: Бонусный урок



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ex-TheQuestion, ex-Ultimate Guitar, ex-
МегаФон

Содержание урока

- ❑ Что дальше?
- ❑ Обзор основных программ
- ❑ Воркшоп 1: Дэшборд в Google Data Studio
- ❑ Воркшоп 2: Дэшборд в Mode Analytics
- ❑ Воркшоп 3: Дэшборд в Tableau
- ❑ Воркшоп 4: Дэшборд в Yandex DataLens



Что дальше?

Решения и методики построения запросов
для разработчиков баз данных

Включает решения
для SQL Server, PostgreSQL,
Oracle, MySQL и DB2



Читать книги и профильные сайты

Practice > Sql

80 r
Rank

Sql

Revising the Select Query I

Easy, Max Score: 10, Success Rate: 95.84%

Query the data for all American cities with populations larger than 100,000.



Solve Challenge

Revising the Select Query II

Easy, Max Score: 10, Success Rate: 98.78%



Solve Challenge

Select All

Easy, Max Score: 10, Success Rate: 99.67%



Solve Challenge

Select By ID

Easy, Max Score: 10, Success Rate: 99.70%



Solve Challenge

Участвовать в Контекстах

Report Share Options View x Describing Retention Acme.io Switch

Describing Retention Query 1 SQL Display Table Area Chart 100% Line Chart 1

Report Layout

QUERIES

- Scratch Query
- Query 1
 - SQL
 - Display Table
 - Area Chart 100%
 - Line Chart 1
 - Add Chart
- Query 3
 - SQL
 - Display Table
 - Area Chart 100%
 - Line Chart 1
 - Add Chart

Run Limit 100

```

01 WITH
02
03 users AS (
04     SELECT customer_id AS user_id,
05            signup_date AS activated_at
06     FROM bern.sample_customer_table
07 ),
08
09 events AS (
10     SELECT customer_id AS user_id,
11            event_name,
12            event_date AS occurred_at
13     FROM bern.sample_event_table
14 ),
15
16 SELECT DATE_TRUNC('week', u.activated_at) AS signup_date,
17
18        TRUNC(CAST('day', e.occurred_at - u.activated_at)/7) AS user_period,
19
20        COUNT(DISTINCT e.user_id) AS retained_users
21 FROM users u
22 JOIN events e
23   ON e.user_id = u.user_id
24  AND e.occurred_at >= u.activated_at
25  AND e.occurred_at <= '2014-10-31':TIMESTAMP
26 WHERE u.activated_at >= '2014-10-31':TIMESTAMP - INTERVAL '11 week'
27      AND u.activated_at <= '2014-10-31':TIMESTAMP
28 GROUP BY 1,2
  
```

Ready

Chart Download CSV 2,205 rows

column_1	column_2	column_3	column_4	column_5	column_6
22,502	4288	4224	4543	7705	13228
5140	10208	6858	18806	4175	13525
8723	6121	15639	472	11653	238
889	209	9544	16209	16875	20193
7477	11887	9904	18292	10267	12866

Database: Acme.io Prod

Filter for tables, columns

PINNED

sfgov.sf_rec_park_evaluation...

sfgov.sf_rec_park_info

TABLES

sfgov.parking_tickets

sfgov.more_parking_tickets

sfgov.film_locations

sfgov.noise_complaints

SFGOV.SF_REC_PARK_EVA... Pin

- sq
- psa
- park_id
- sq
- id

**Практиковаться
на открытых данных**

Обзор основных программ

ВІ інструменти



BI-платформы (business intelligence) собирают данные из различных источников, строят регулярные отчеты и интерактивные дашборды.

Используются для создания систем аналитической отчетности, мониторинга, KPI и отвечают на вопросы: что случилось ранее или происходит в текущий момент.

Tableau

- ❑ На рынке с 2003 года.
- ❑ Очень быстро работает данными.
- ❑ Продвинутая визуализация.
- ❑ Высокая стоимость.



Tableau

Connect

To a file

Excel

Text File

Statistical File

Other files

To a server

Tableau Server

QuickBooks

Google Sheets

Microsoft SQL Server

MySQL

More Servers... >

Saved data sources

Sample - Superstore

World Indicators

Search

Tableau Server

Amazon Aurora

Amazon EMR

Amazon Redshift

Cloudera Hadoop

EXASolution

Firebird

Google Analytics

Google BigQuery

Google Cloud SQL

Google Sheets

Hortonworks Hadoop Hive

HP Vertica

Kognitio

MapR Hadoop Hive

Microsoft SQL Server

MySQL

OData

Oracle

Pivotal Greenplum Database

PostgreSQL

QuickBooks

Salesforce

SAP HANA

Snowflake

Spark SQL

Teradata

Web Data Connector

Connect to Google Sheets

Search by sheet title or by URL (ex: https://docs.google.com/spreadsheets/d/{GoogleSheetsID})

Search

Name	Owned by	Last opened by me
Zygon Sightings	Me	11:05 AM
Disease Data	Me	11:22 AM
Cybermen sightings_Antartica	UNIT	Dec 3, 2015
Cybermen sightings_Australia	UNIT	Dec 3, 2015
Cybermen sightings_China	UNIT	Dec 3, 2015
Cybermen sightings_Europe	UNIT	Dec 3, 2015
Cybermen sightings_India	UNIT	Dec 3, 2015
Cybermen sightings_Middle East	UNIT	Dec 3, 2015
Cybermen sightings_North America	UNIT	Dec 3, 2015
Cybermen sightings_South America	UNIT	Dec 3, 2015
Foretold_Accounts	Professor Moorhouse	Jun 8, 2014
Library checkout	River Song	Sep 9, 2008
Map anomalies	River Song	Nov 9, 2006
Map anomalies_v2	Me	Nov 9, 2003
Oswald_Oswin_impossible	Me	Sep 2, 2000

Details

Map/Worksheet: Number of Sightings

Country	Sightings
USA	10
UK	10
France	10
Italy	10
Germany	10
Spain	10
Sweden	10
Belgium	10
Denmark	10
Poland	10
Portugal	10
Japan	10
Australia	10
Canada	10
South Africa	10
India	10
China	10
South America	10
Antarctica	10

Zygon Sightings

Last Modified on **March 12, 2016**

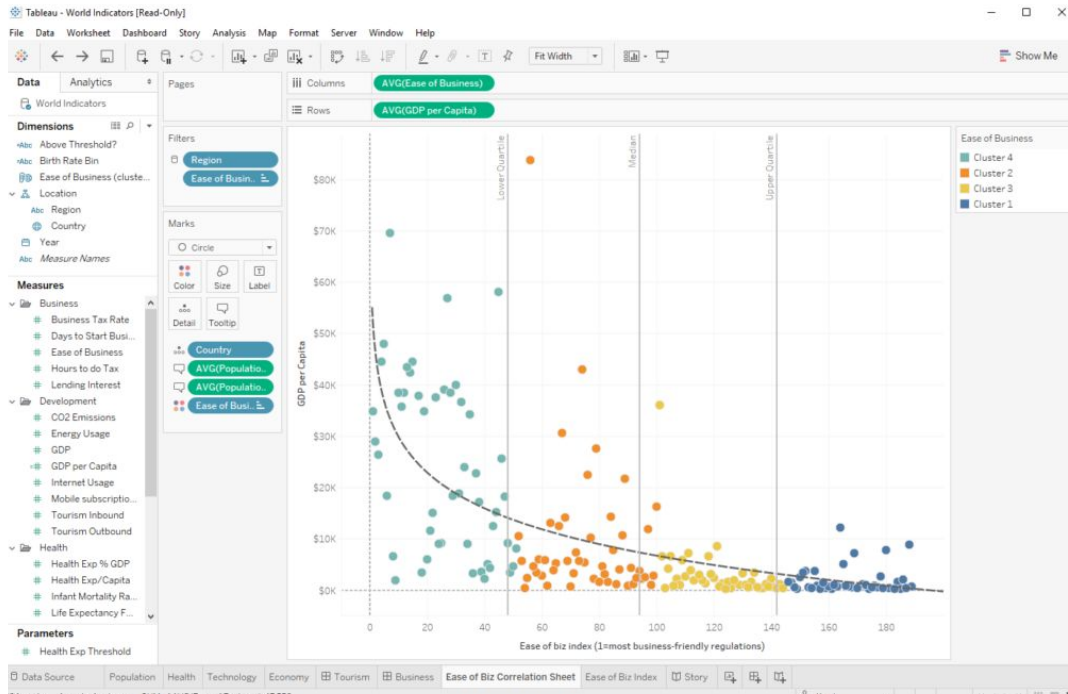
Last Modified by **Me**

[Open in Google Sheets](#)

Cancel

Connect

Tableau



Tableau

Data Sources

sales+ (Multiple Connections)

Connections

Sales

Oracle

Store

Microsoft SQL Server

Population

Excel

Add

Sheets

Enter sheet name

lowa

New Union

Sort Fields

Data source order

Customers Zip Code	Customers Address	Customers Name	Customers Store1	lowa zip code (lowa)	lowa Population	Products Item No
50613	2217 Colleg...	Hillstreet News an...	2,106	50613	42,031	11,297
50613	2217 Colleg...	Hillstreet News an...	2,106	50613	42,031	25,607
50613	2217 Colleg...	Hillstreet News an...	2,106	50613	42,031	27,554
50613	2217 Colleg...	Hillstreet News an...	2,106	50613	42,031	28,867
50613	2217 Colleg...	Hillstreet News an...	2,106	50613	42,031	40,965
50613	2217 Colleg...	Hillstreet News an...	2,106	50613	42,031	40,976
50613	2217 Colleg...	Hillstreet News an...	2,106	50613	42,031	43,336

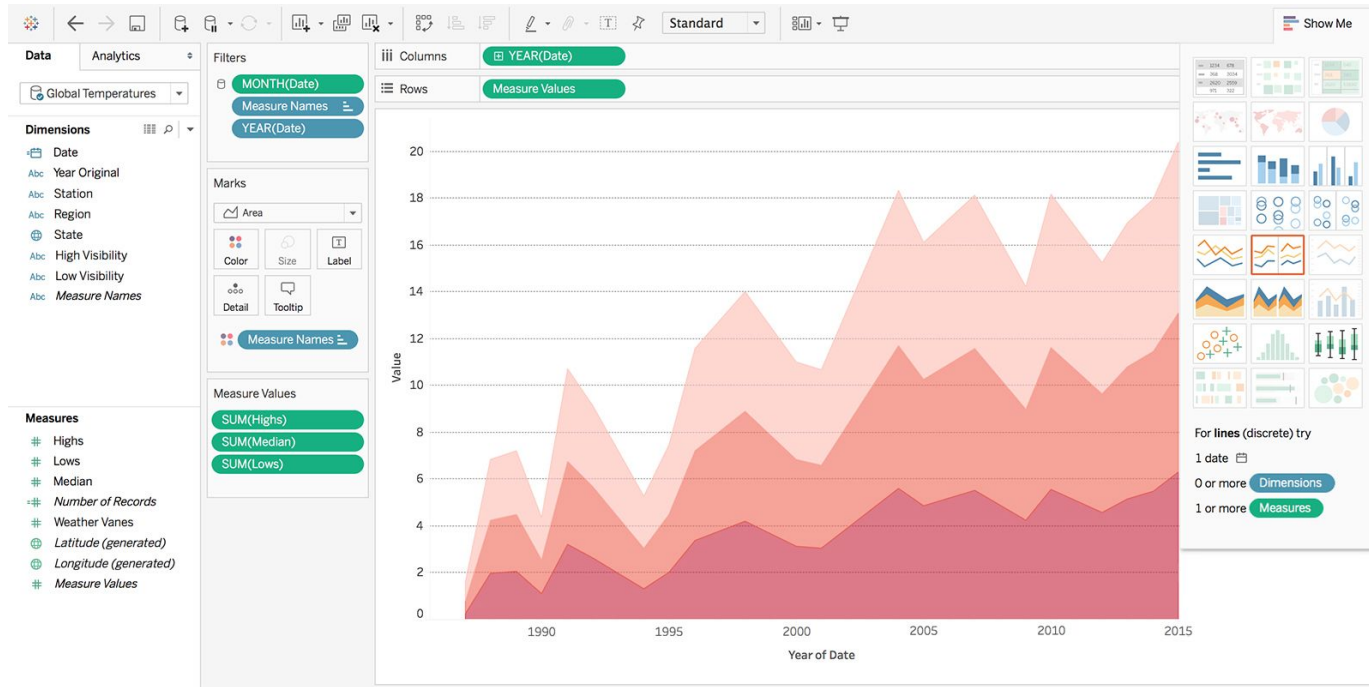
Sales (Oracle)

Customers (SQL Server)

Population (Excel)

Products (SQL Server)

Tableau



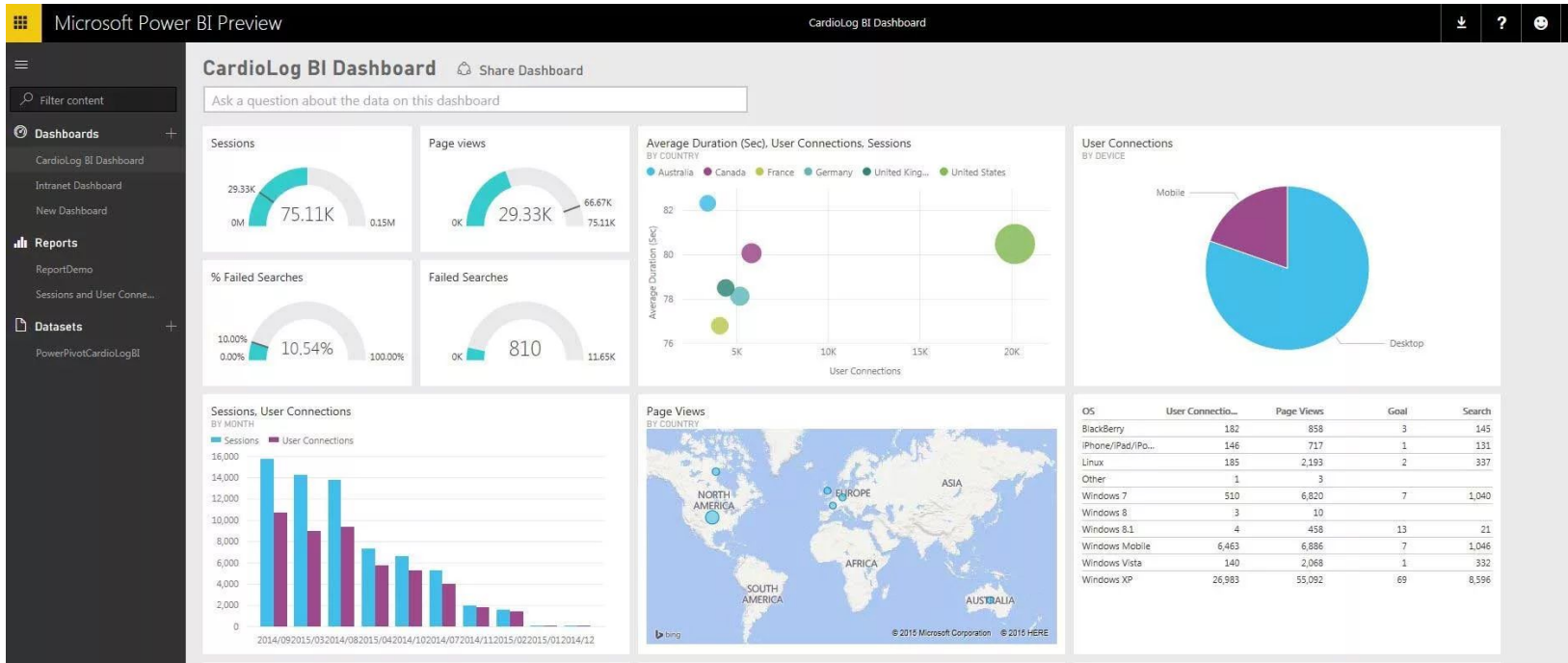
PowerBI

- ❑ Enterprise решение.
- ❑ Тесно связан с сервисами Microsoft Office.

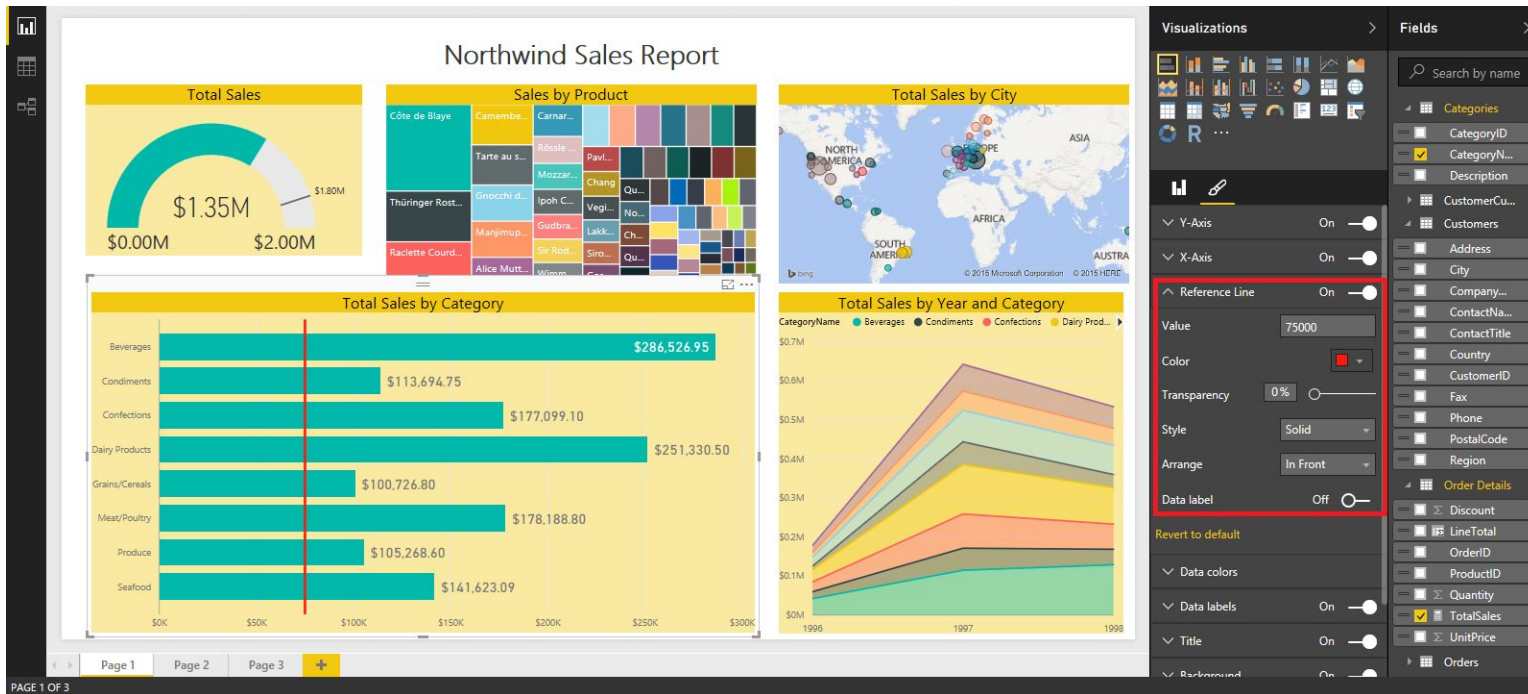


Power BI

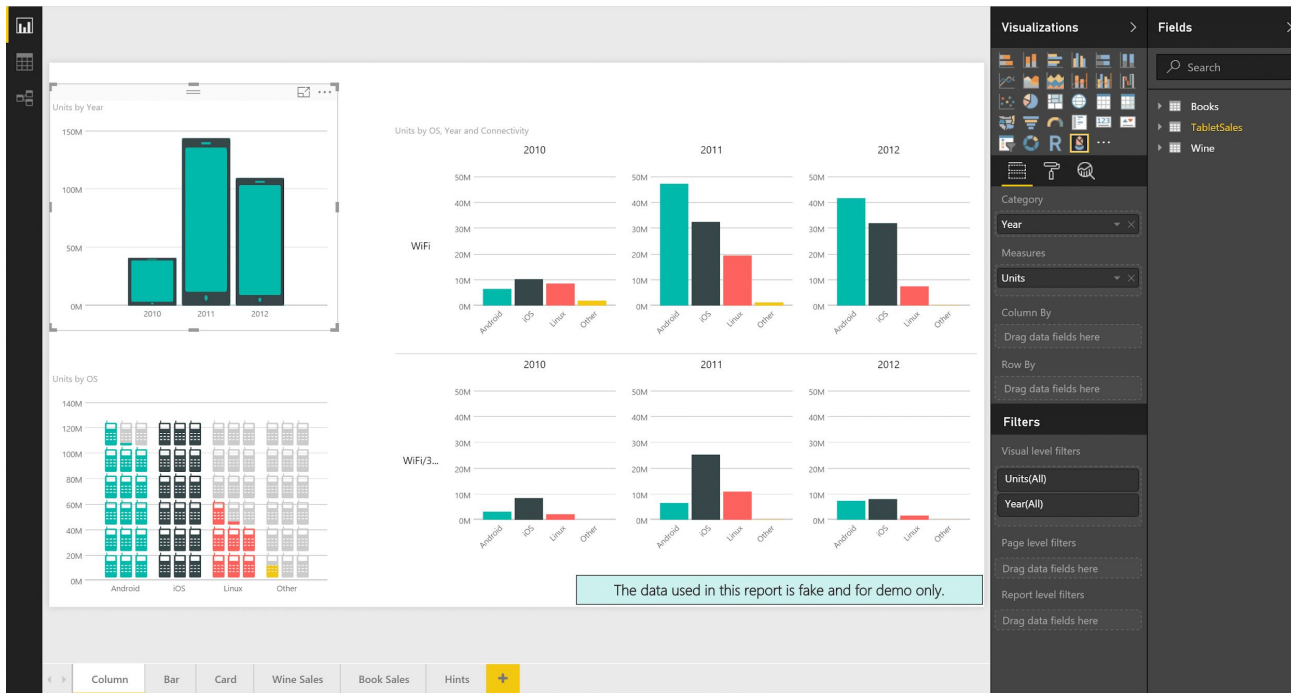
PowerBI



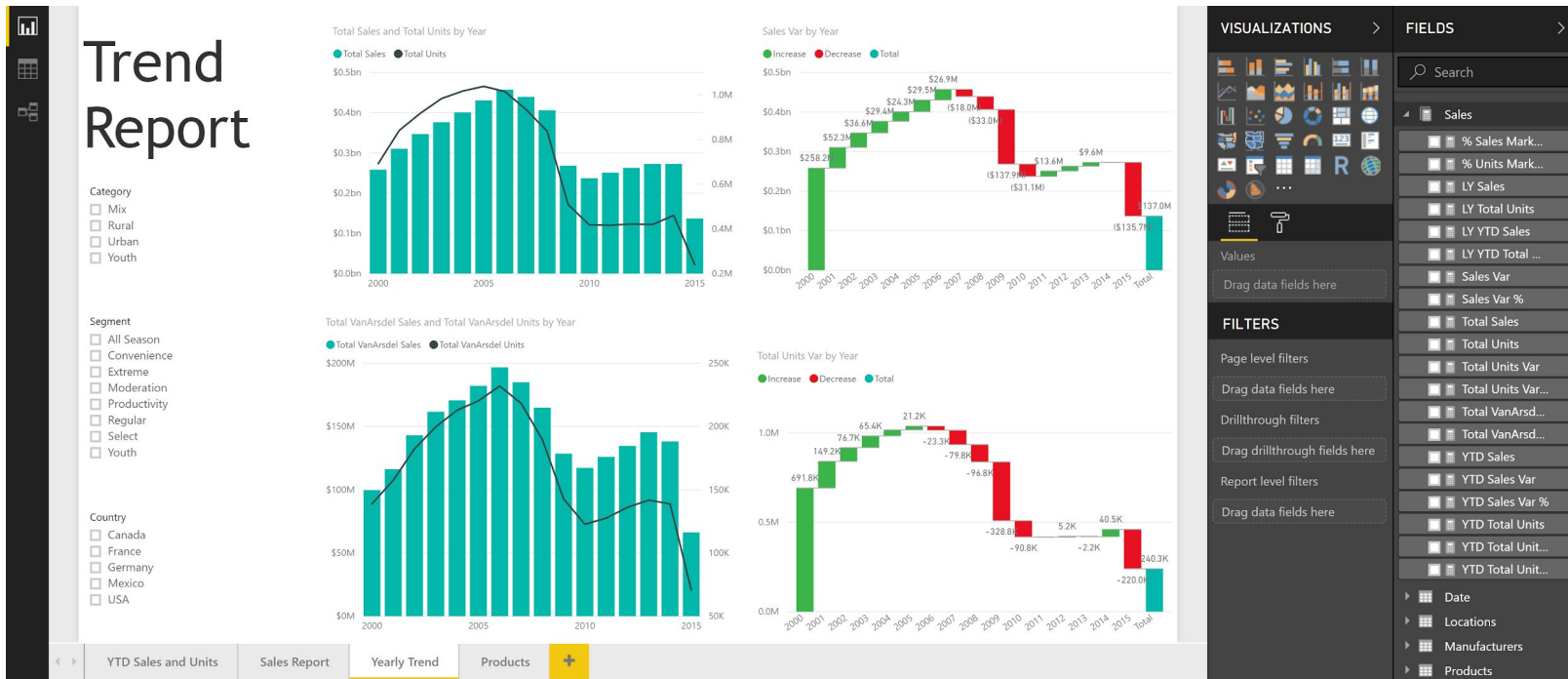
PowerBI



PowerBI



PowerBI



Google Data Studio

- ❑ Платить не обязательно :)
- ❑ Хорошо настроена связь с сервисами Google.



Google
Data Studio

Google Data Studio

 Search

DEVELOPERS

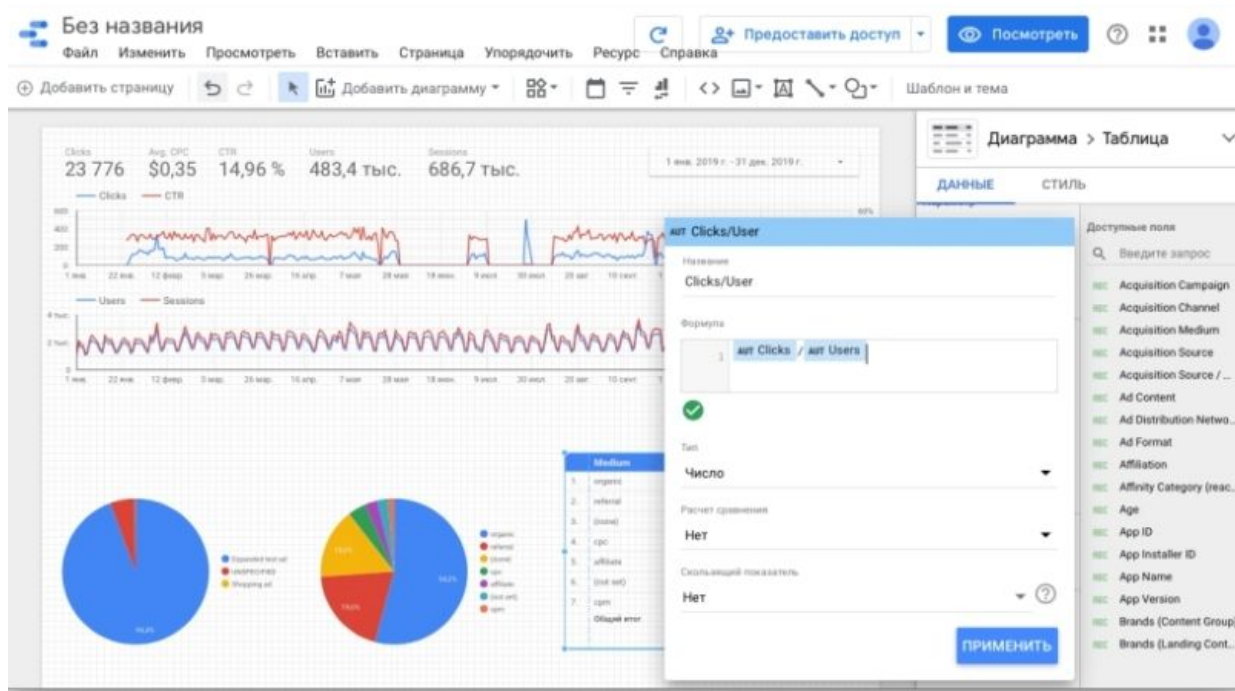
Google connectors (18)
Connectors built and supported by Data Studio. [Learn more](#)

 Google Analytics By Google Connect to Google Analytics reporting views. Learn more	 Google Ads By Google Connect to Google Ads performance report data. Learn more	 Google Sheets By Google Connect to Google Sheets. Learn more	 BigQuery By Google Connect to BigQuery tables and custom queries. Learn more	 File Upload By Google Connect to CSV (comma-separated values) files. Learn more
 Campaign Manager By Google Connect to Campaign Manager data. Learn more	 Cloud Spanner By Google Connect to Google Cloud Spanner databases.	 Cloud SQL for MySQL By Google Connect to Google Cloud SQL for MySQL databases. Learn more	 Display & Video 360 By Google Connect to Display & Video 360 report data.	 Extract Data By Google Connect to Extract Data Learn more
 Google Ad Manager By Google Connect to Google Ad Manager data. Learn more	 Google Cloud Storage By Google See your files in Google Cloud Storage. Learn more	 MySQL By Google Connect to MySQL databases. Learn more	 PostgreSQL By Google Connect to PostgreSQL databases. Learn more	 Search Ads 360 By Google Connect to Search Ads 360 performance reports.
 Search console By Google Connect to Search Console data. Learn more	 YouTube Analytics By Google Connect to YouTube Analytics data. Learn more	 TV Attribution By Google Connect to TV Attribution data. Learn more		

Google Data Studio



Google Data Studio

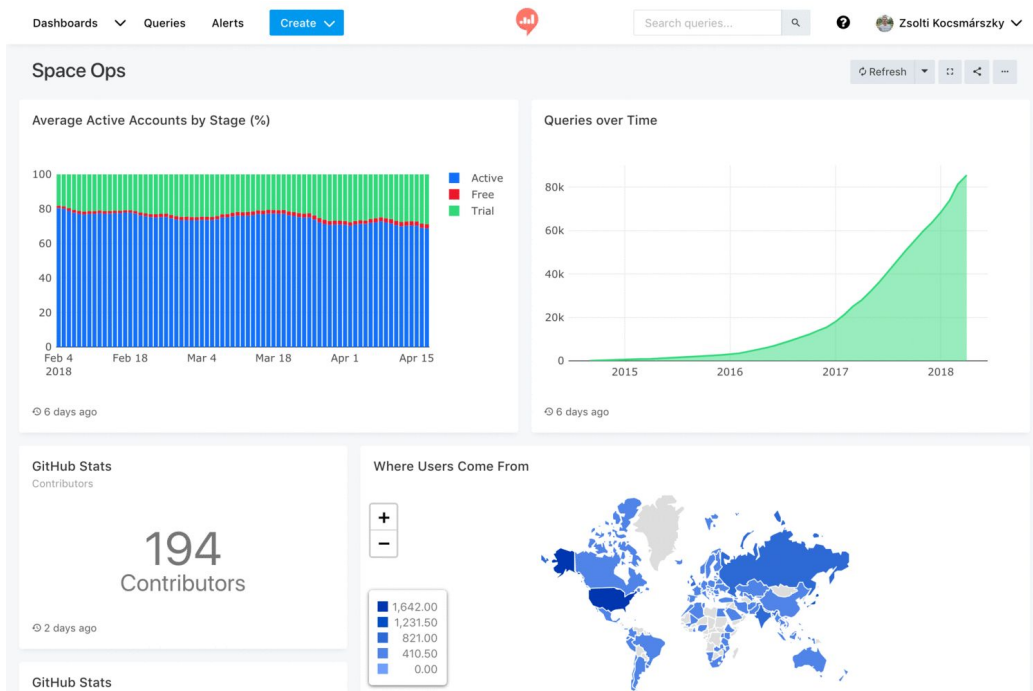


Redash

- ❑ Есть триальный период (1 месяц).
- ❑ Настроены интеграции с популярными системами аналитики и хранилищами.



Redash



Metabase

- ❑ Есть бесплатная (опенсорс) и платная версия.
- ❑ Можно настраивать рассылки и оповещения.
- ❑ Можно делать дэшборды как на основе SQL запросов, так и при помощи интерфейсного выбора полей.

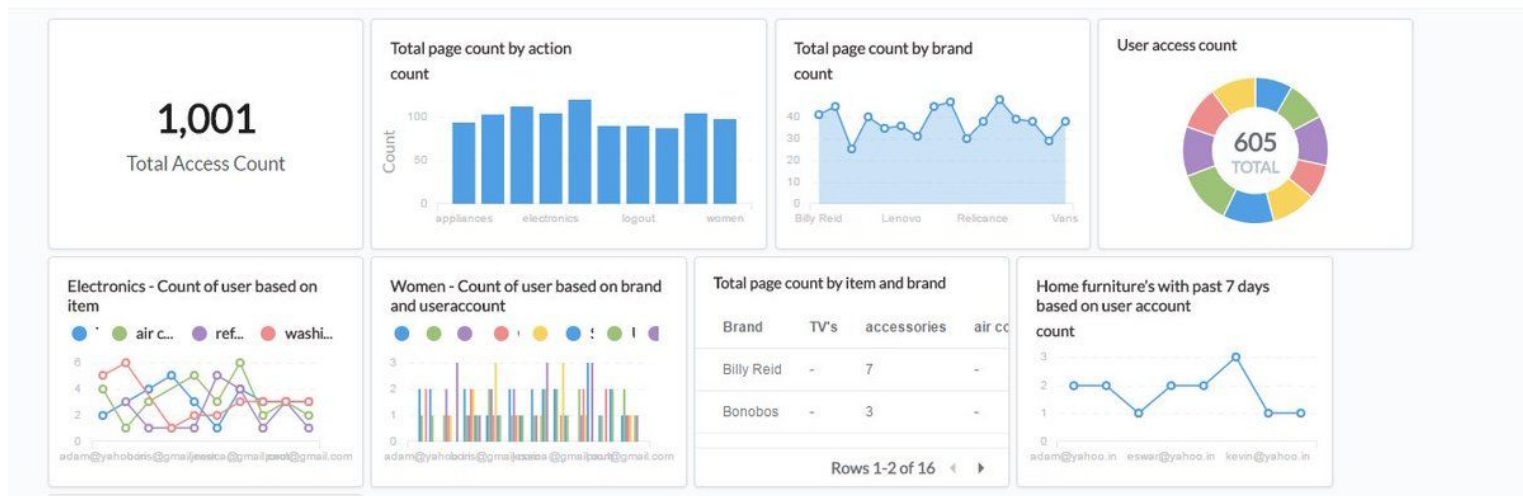


Metabase

Metabase



user analysis
No description yet



Metabase



Dashboards ▾

Questions

Pulses

Data Reference

New Question

AS ▾

Zone Wise Cluster Count

DATABASE

Prod-Replica ▾

HIDE EDITOR

```
1 select
2   z.description,
3   count(*)
4 from clusters as c
5 join zones as z
6 on c.zone_id = z.id
7 where {{city}}
8 group by z.description |
```

VISUALIZATION

Table ▾



Refresh

Showing 51 rows. 

description	Count
Faridabad	7
Velachery	6
Ambattur Industrial area	5
T Nagar	6

Variables

SETTINGS

HELP

city

Filter label

City

Variable type

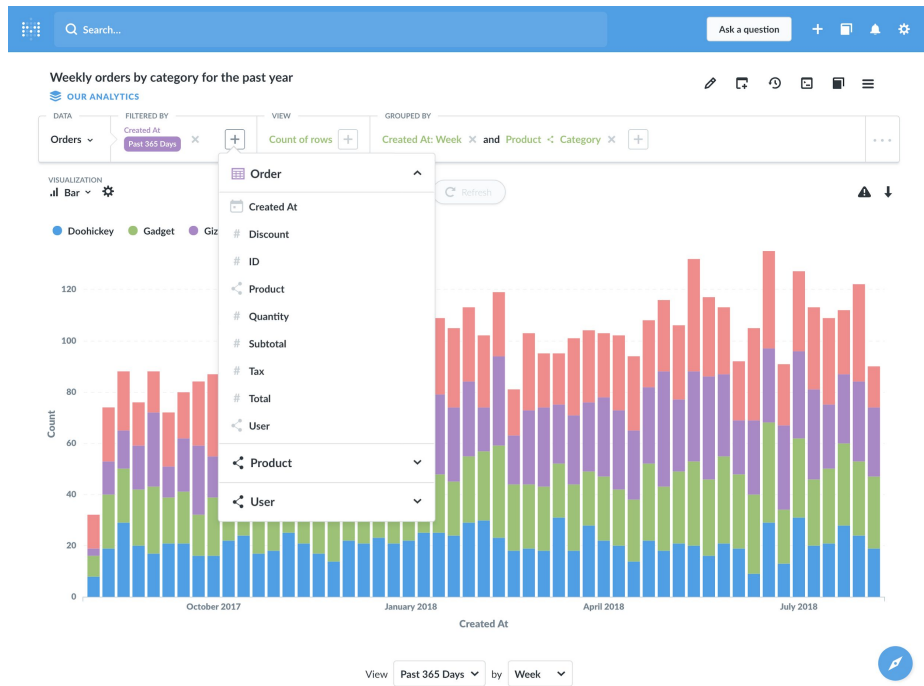
Field Filter ▾

Field

PUBLIC > CLUSTERS

Geo Region ID ▾

Metabase

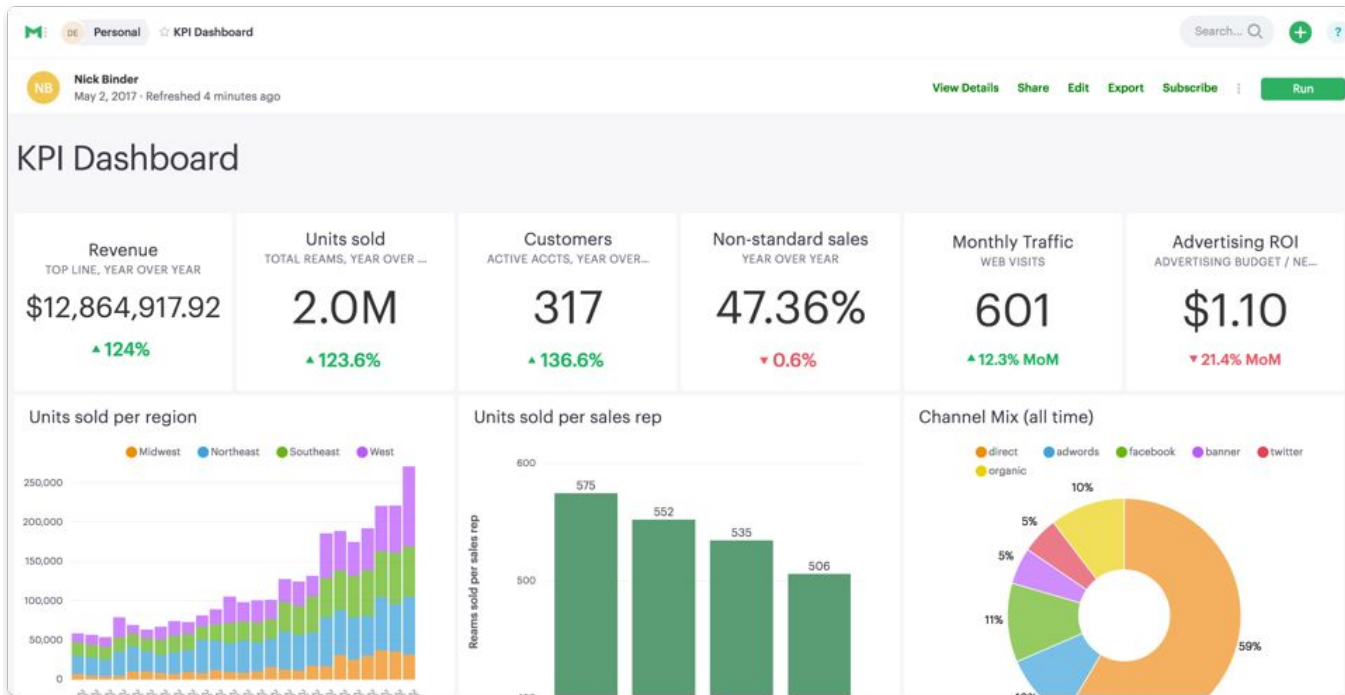


Mode

- ❑ Можно строить дэшборды в связке SQL + Python.
- ❑ Есть бесплатный доступ до тестовых таблиц.



Mode



Mode

DE Community ☆ Sample - KPI dashboard

Search... Q



Nick Binder

April 28, 2017 - Refreshed 4 minutes ago

[View Details](#)

[Share](#)

[Clone](#)

[Edit](#)

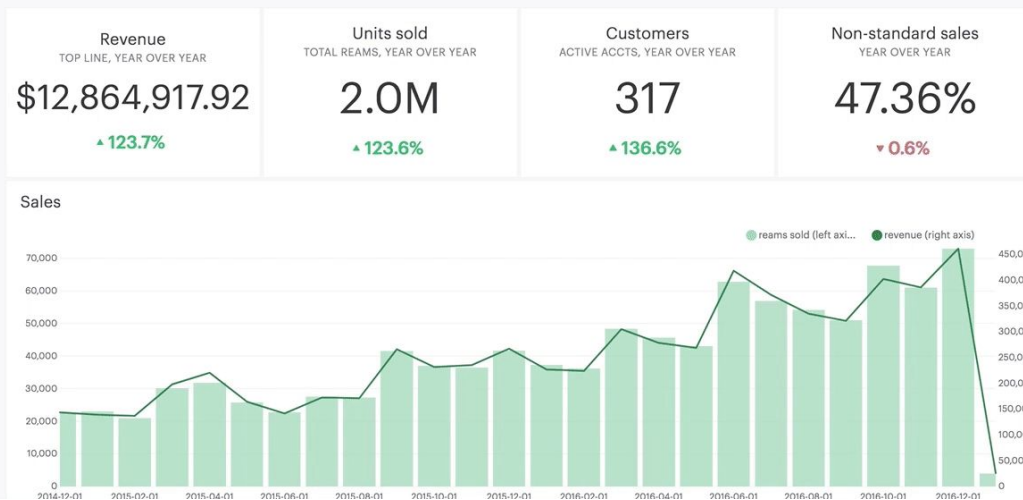
[Fullscreen](#)

[Open Filters](#)

[Run](#)



Sample - KPI dashboard



Mode

The screenshot displays the Mode SQL interface. At the top, the header shows the user 'Marissa Gemma' and the report title 'Demo report', with options to 'Report', 'Embed', 'Share', 'Schedule', and 'View'. Below the header, a toolbar includes a 'Run' button, a 'Limit 100' checkbox, and buttons for 'Format SQL' and 'View History...'. An orange arrow points from the 'Format SQL' button to the SQL query editor. The query editor contains the following SQL code:

```
1 SELECT *  
2 FROM tutorial.us_housing_units
```

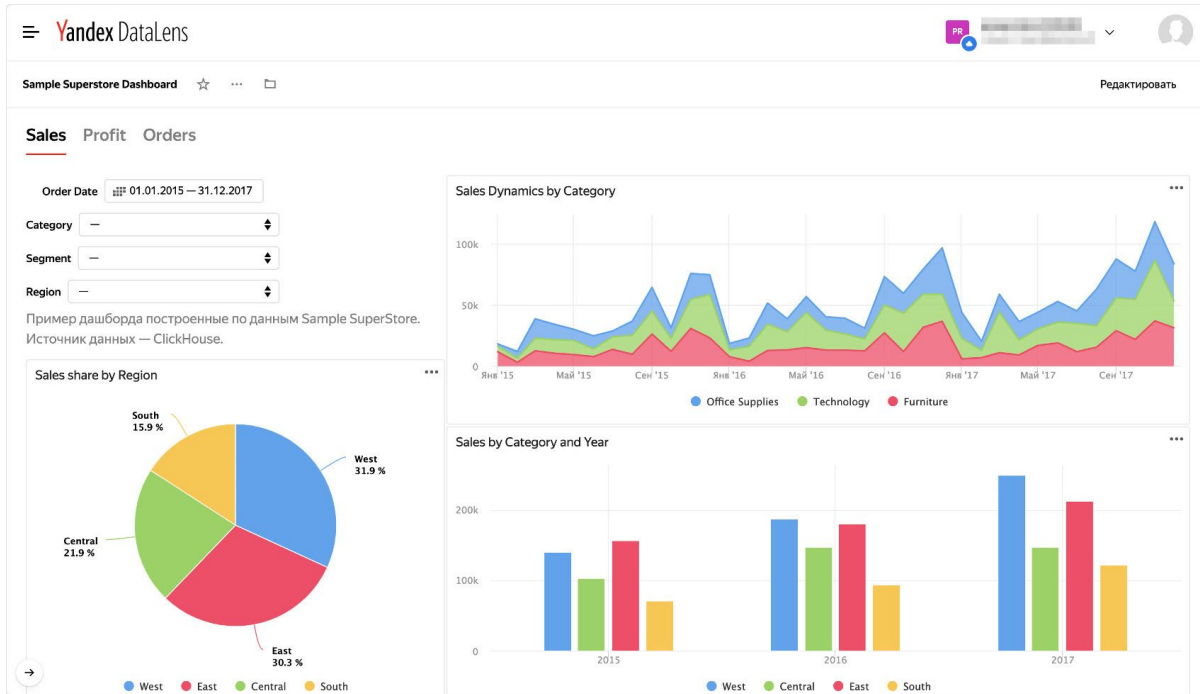
Below the query editor, a status bar shows a green checkmark and the text 'Succeeded in 941ms'. An orange arrow points from this status bar to the bottom of the interface. The bottom of the interface features buttons for 'Export', 'Copy', 'Chart', and 'Pivot', followed by the text '559 rows returned' and icons for table and chart views. On the left side, a sidebar lists navigation options: 'REPORTING' (with 'Report Builder' checked), 'NOTEBOOK' (with 'Python Notebook'), and 'QUERIES' (with 'Query 1' expanded and 'SQL' selected). On the right side, a panel for 'Mode Public Warehouse' shows 'Tables' and 'Definitions' tabs, a search bar, and a list of pinned tables including 'tutorial.us_housing_units'.

DataLens

- ❑ Поддерживает многие источники и форматы.
- ❑ Есть возможность делать публичные дэшборды.
- ❑ В бесплатной версии есть открытые данные.

Yandex DataLens

Datalens



DataLens

Консоль

DataLens

Все объекты

Избранное

Подключения

Датасеты

Чарты

Дашборды

Marketplace

Документация

Поддержка

Тарифы

Демонстрационный режим

Для полноценной работы необходимо создать платёжный аккаунт Яндекс.Облака. Мы не спишем деньги, а вы сможете работать со своими данными в бесплатном тарифе DataLens.

[Создать платёжный аккаунт](#)



Подключения

Подключайте свои источники данных

[Создать подключение](#)



Датасеты

Формируйте наборы данных с вычисляемыми полями и агрегациями

[Создать датасет](#)



Чарты

Визуализируйте данные в виде диаграмм и таблиц

[Создать чарт](#)



Дашборды

Создавайте страницы с наборами диаграмм, таблиц и фильтров

[Создать дашборд](#)

Примеры



Дашборд по данным Метрики



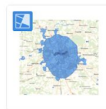
Дашборд по данным из ClickHouse



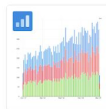
Круговая диаграмма



Точечная диаграмма



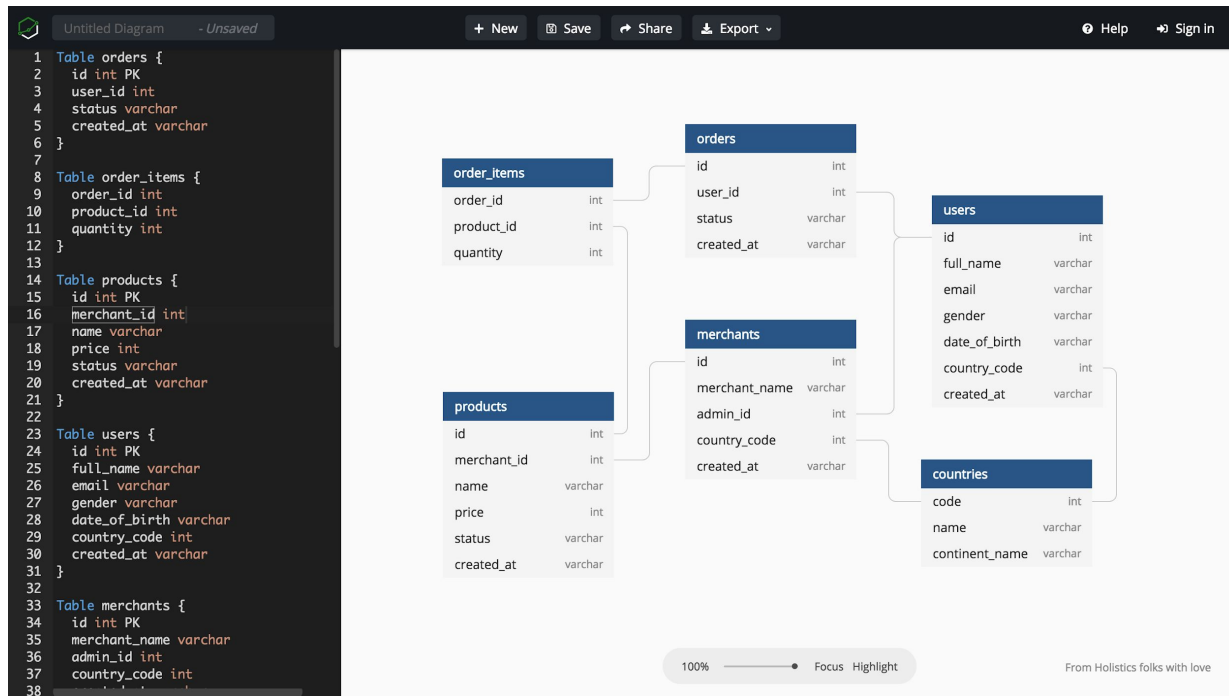
Фоновая карта



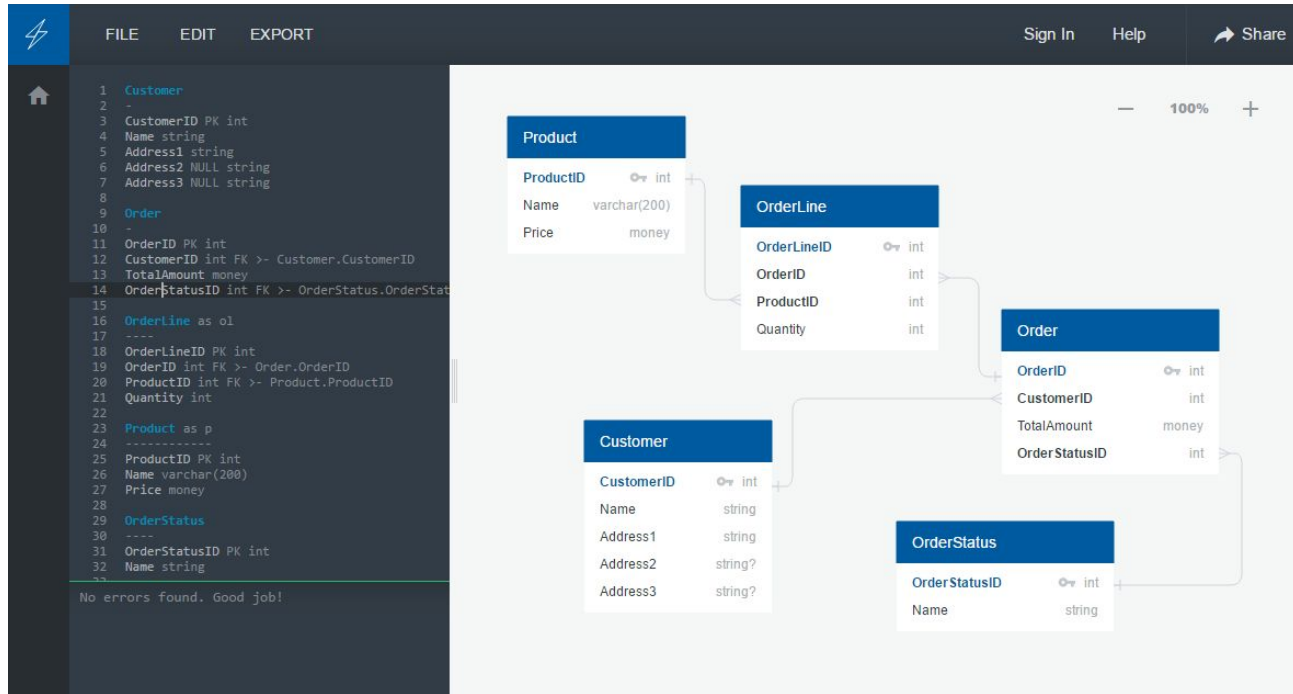
Столбчатая диаграмма

Проектирование баз данных

dbdiagram.io



QuickDBD



Draw.io

